

JEE MAIN REVISION MASTER: MATHEMATICAL INDUCTION OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Mathematical Induction - Core Application	3 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mathematical Induction - Core Application (Moderate Yield — 3 PYQs)

Core Principle: Key problem pattern in Mathematical Induction.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2005 Shift] Q#1 (MCQ): "If $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}$, then which one of the followin..." [Ans: N/A]
- [2004 Shift] Q#2 (MCQ): "Let $S(K) = 1+3+5...+(2K-1) = 3+K^2$. Then which of the following is true..." [Ans: N/A]
- [2002 Shift] Q#3 (MCQ): "If $a_n = \sqrt{(7+\sqrt{7}+\sqrt{7}+.....)}$ having n radical signs then by methods of mathematical induction which is true..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: AIEEE 2005 (Q#1)

If $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction?

- (A) $A^n = nA - (n-1)I$
- (B) $A^n = 2^n - 1A - (n-1)I$
- (C) $A^n = nA + (n-1)I$
- (D) $A^n = 2^n - 1A + (n-1)I$

Correct Answer: None

Solution Analysis: Given $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ $\therefore A \times A = A^2 = \begin{bmatrix} 1 & 0 & 2 & 1 \end{bmatrix}$ and $A^3 = A^2 \times A = \begin{bmatrix} 1 & 0 & 3 & 1 \end{bmatrix}$ So we can say $A^n = \begin{bmatrix} 1 & 0 & n & 1 \end{bmatrix}$ Now $nA - (n-1)I = \begin{bmatrix} n & 0 & n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 & 0 & n-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & n & 1 \end{bmatrix} = A^n \therefore A^n = nA - (n-1)I$